## Referee cover memo

2 November 2026

**Re:** *Quotient-Group Obstructions for Pure Type II Divisor Constructions in the Erdős–Straus Equation*

The accompanying five-page note extracts a limited classification problem from an AI-assisted investigation. It is submitted for informal mathematical and bibliographic assessment, not as a claim to resolve the Erdős–Straus conjecture. The manuscript is self-contained; the historical ledger is not needed to read its arguments.

**What is established.** For $x_m = q_1^m q_2^r q_3^s$ at a fixed residual modulus $a \equiv 3 \pmod 4$, the note specifies three simultaneous requirements: a Type II square-divisor witness, absence of ordinary Type II witnesses, and absence of all Type I square-divisor witnesses. It proves an eventual quotient criterion, the necessary quotient index $|G/\langle q_1\rangle| \geq 8$, and the exact index-eight configuration when $q_1 = 2$. Explicit modular families attain the sharp index bound and the four stated modulus minima.

**Proof versus finite verification.** The quotient criterion, index obstruction, equality classification, and lower bounds for moduli 11 and 31 have complete elementary proofs. The lower bounds 127 for growing base 2 and 195 for growing base 2 with composite modulus use exhaustive integer certificates: all 31 eligible smaller moduli in the first case and all 25 eligible smaller composite moduli in the second. The index obstruction excludes every choice of the remaining primes and exponents at those moduli. The appendix records all arithmetic and supplies standard-library Python code. Finite samples of prime inputs are explicitly separated from the all-exponent modular proofs.

**Why review may be worthwhile.** The divisor parametrization and quotient-by-saturation mechanism are established mathematics or elementary consequences thereof. The potential contribution is the sharp small-quotient exclusion together with its precisely constrained arithmetic realizations. A targeted literature review did not identify an equivalent statement with the same ordinary-versus-quadratic exclusions. **Mathematical novelty remains unconfirmed**; absence of an exact match is not evidence of publication priority or significance.

**Principal question for the referee.** Is the sharp index-eight classification, with these specific purity conditions and modulus minima, already a direct published consequence of known divisor or finite-group results; if not, does it constitute a useful observation worth recording?

The note makes no assertion of infinitely many prime values, prime-product minimality, or four-dimensional minima. No withdrawn paper is used as reliable support. The final check corrected two draft coordinate transcriptions and documented a remaining ledger typo separately: the modulus-195, $m = 8$ example has $y = 65{,}001{,}584$. The master ledger was not changed.

**Editorial verdict: Ready for informal expert review.**